

Homework 3 – Physical Oceanography

Total 10 points

Due date: Tuesday 07/19/2022

- 1- Briefly explain how and where deep waters form and how that process influences global circulation? (2 Pts)**
- 2- Answer the following questions about ocean Gyres (2 Pts)**
 - a. What are the main forces (apparent and real) involved in the spin-up of oceanic gyres by the winds? (0.5 Pt) (Answer: friction, pressure gradient, and Coriolis. If students mention these 3 and gravity give them full grade).**
 - b. Describe the chain of events of how winds accelerates the ocean gyres in the Northern Hemisphere. You can create your own schematics to aid your explanation. (1.5 Pts).**
- 3- What is coastal upwelling? How does it form? What is its relation to local weather and fisheries (2 pts)**
- 4- Consider a surface gravity waves with wavelength of 100 m (please indicate your reasoning in each item) (2 pts)**
 - a. What is its propagation speed in the deep ocean?**
 - b. Once it reaches regions with local depth of approximately 5 m, these waves become shallow water waves. What are the new propagation speeds over this depth?**
 - c. Explain why they slow down or speed up.**
 - d. Why shallow water waves tend to align with local topography and become parallel to the coastlines?**

5- The Bay of Fundy (https://www.youtube.com/watch?v=EnDJ6_XpGfo) presents the largest tidal amplitudes of the world. Read Holtz (2014) (<https://www.alternativesjournal.ca/climate-change/fundy-rising/>) and answer (2 pts)

- a. Why do coastal areas have distinct tidal amplitudes and characteristics?
- b. What forces the tides and why tides can be considered a renewable energy source?
- c. What are the challenges of harnessing tidal energy in the Bay of Fundy?